



Off by One

Ontario Math C3.2 — read, find, and fix code

Name: _____

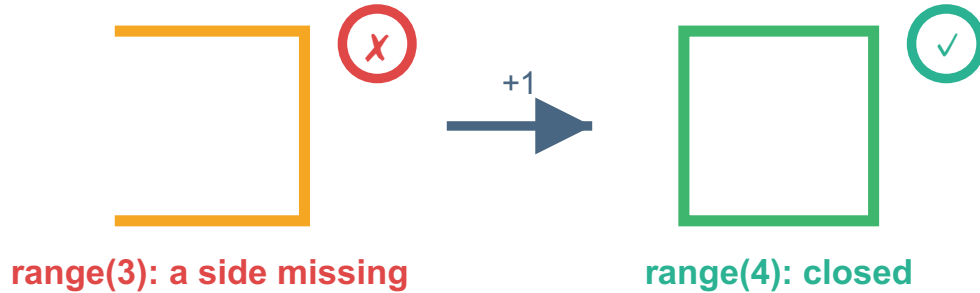
Date: _____

★ I can fix a loop whose repeat number is one too many or too few.

One too many or one too few

A very common bug is being OFF BY ONE — the repeat number is just one more or one less than it should be.

In a turtle shape, off by one is easy to see: a square needs `range(4)`; `range(3)` leaves a side missing.



`range(3)` leaves the square open; `range(4)` closes it.

1 Too many

Bit should repeat 5 times, but the code says repeat 6. How many too many is that, and what should it be? _____

2 Too few

Bit should repeat 6 times, but the code says repeat 5. Is it too many or too few? What should it be?

3 Fix the turtle

This should draw a square: `for i in range(5): bit.forward(50); bit.right(90)`. The range is off by one. What should `range()` be?

4 Challenge

Bit's square code is `for i in range(3): bit.forward(50); bit.right(90)`. What does Bit draw now, and what one change fixes it? Explain why.

Big idea

Off-by-one is a tiny bug with a big effect. Always check the repeat number against the goal — one more or one less changes the whole result.